

27th-09-2026

International Cybersecurity Of Digital Forensic Academy (ICDFA)

PROGRAMME: Advance Critical Infrastructure Security (ACIS)

Prepared by: Aminu Ibrahim Jalo
Course: ACICP301-INTRO TO CRITICAL INFRASTRUCTURE
FOR IT LEARNERS

Tittle: BUILDING YOUR FIRST PLC PROGRAM

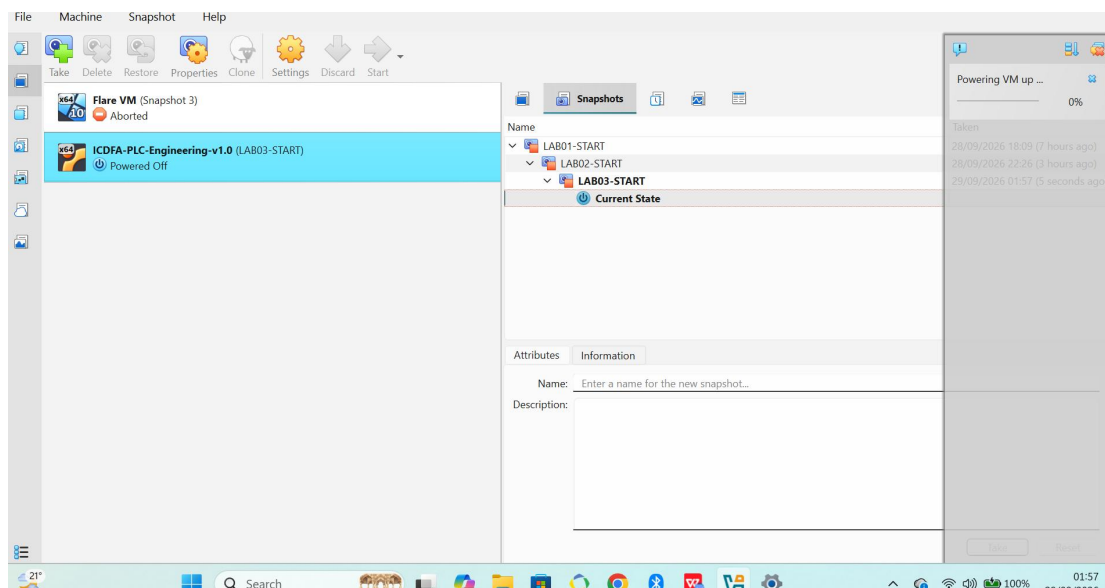
Assignment: Week 1-Lab 2 Report
C11/26/ACIS/17359

Instructor : Aminu Idris

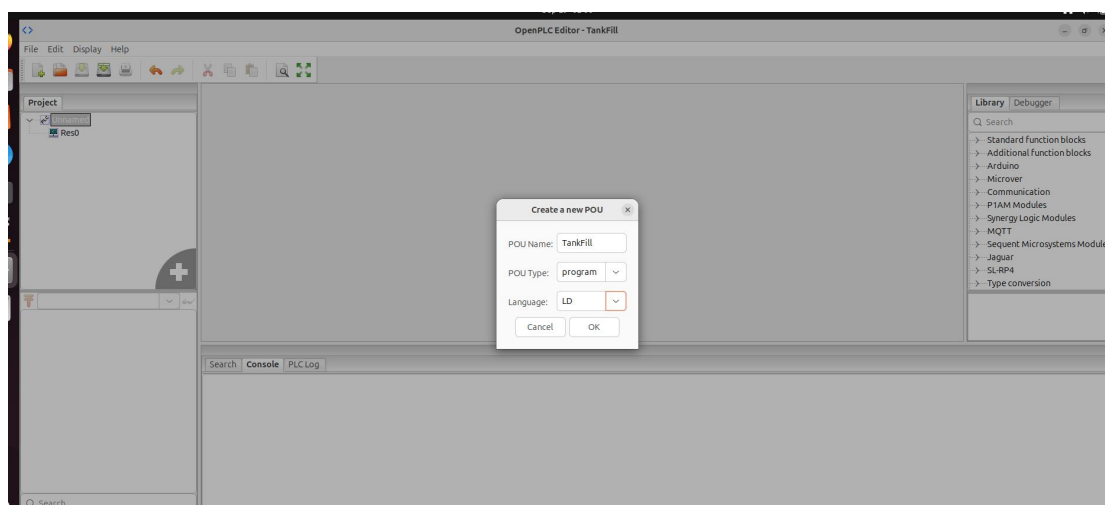
Student Name	Aminu Ibrahim Jalo
Registration Number	C11/26/ACIS/17359
Signature	
Date	27 th -09-2026

Procedure

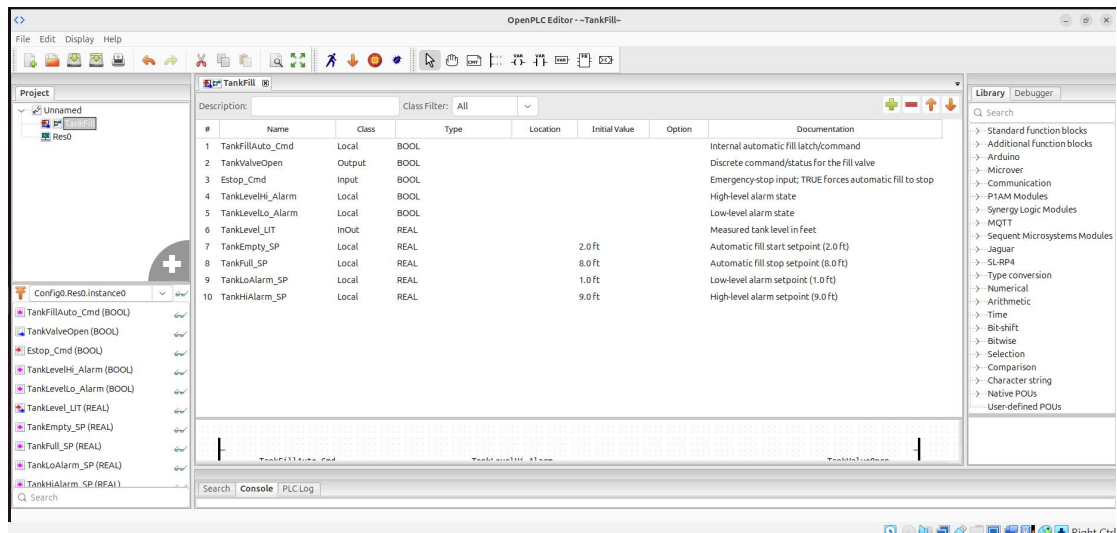
Step 1. Start the PLC VM from a clean snapshot. Launch OpenPLC Editor and create a new project folder named TankFill.



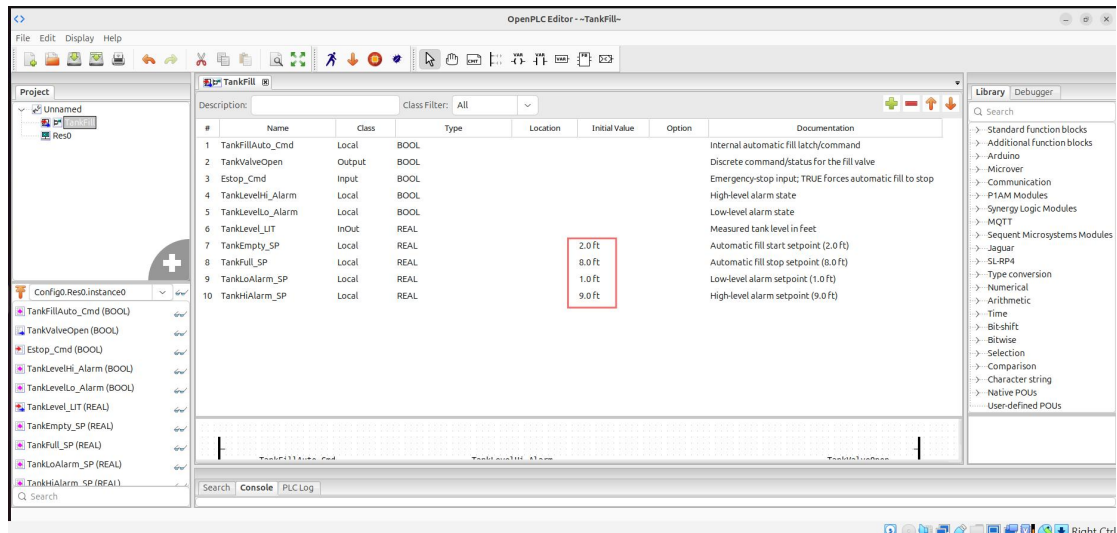
Step 2. Create a new POU named TankFill and select LD (Ladder Diagram) as the language.



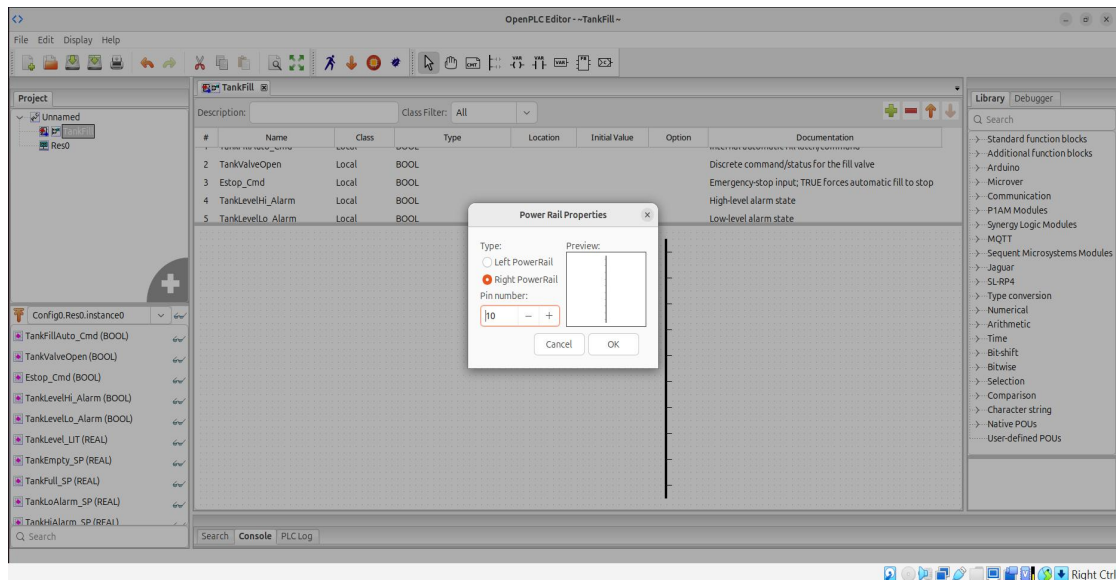
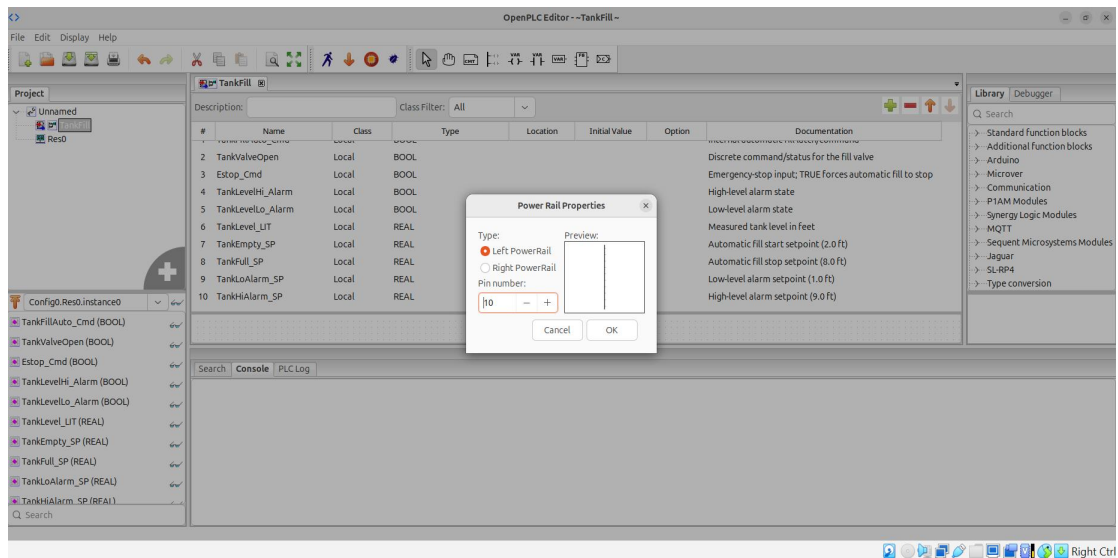
Step 3. Add the variables listed in the Variable Plan table in this manual. Use BOOL for discrete states/commands and REAL for tank-level values and setpoints.



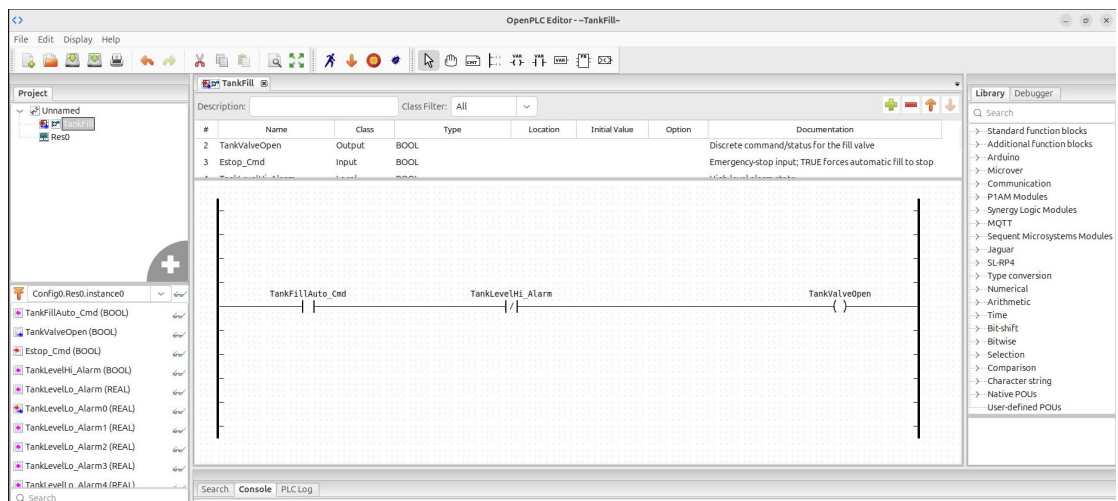
Step 4. Set the operating values: TankEmpty_SP = 2.0 ft, TankFull_SP = 8.0 ft, TankLoAlarm_SP = 1.0 ft and TankHiAlarm_SP = 9.0 ft.



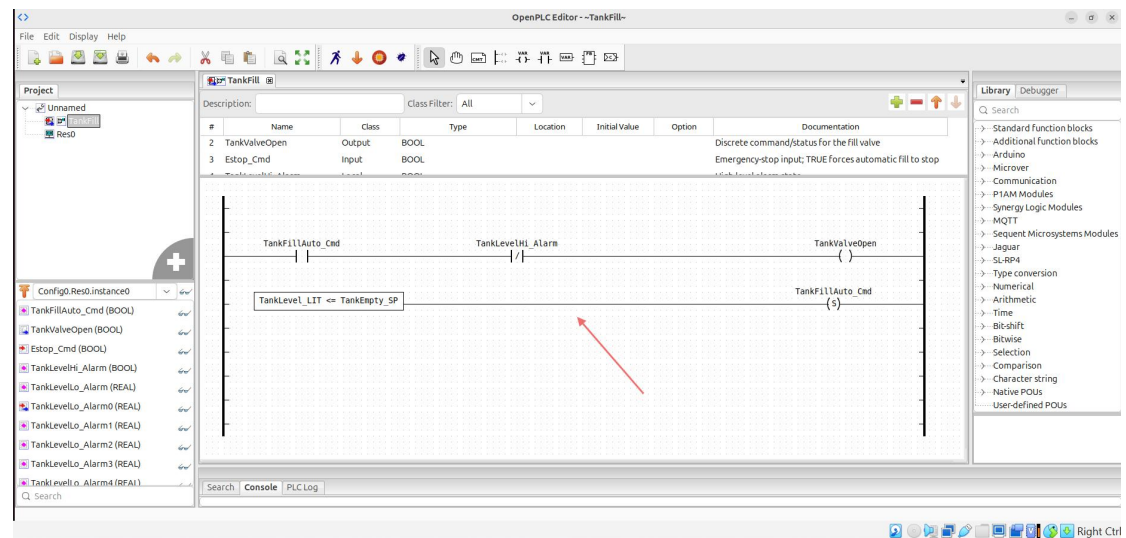
Step 5. Insert left and right power rails with enough pins for the complete program, then create the first logic rung.



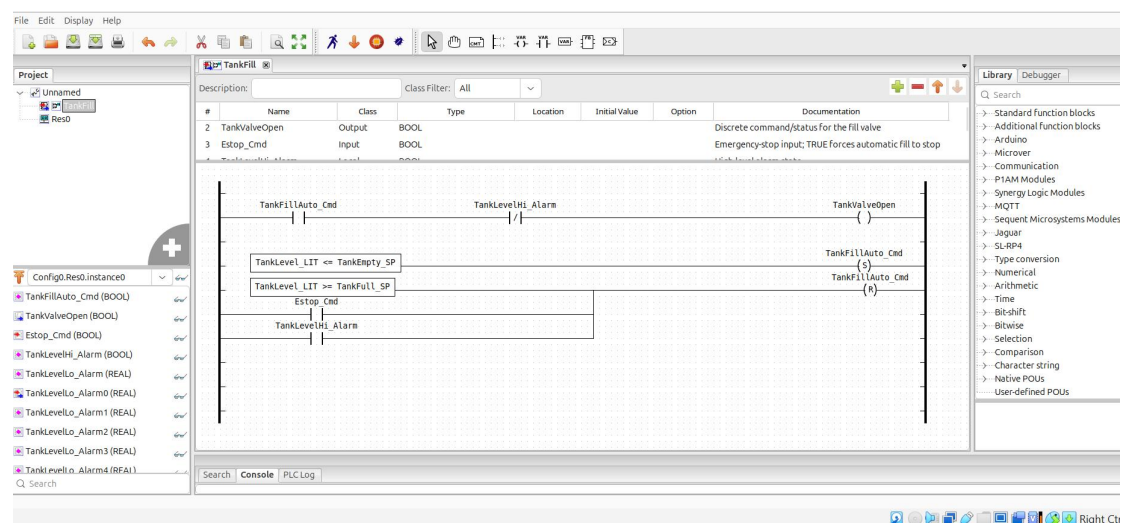
Step 6. Build the valve-control rung: TankFillAuto_Cmd AND NOT TankLevelHi_Alarm energizes TankValveOpen.



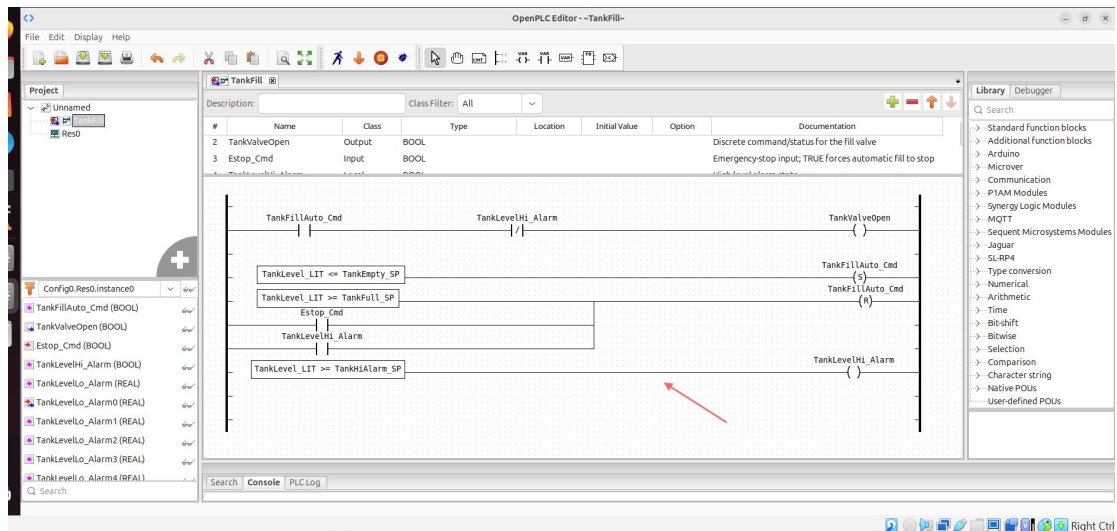
Step 7. Build the automatic-start rung: when TankLevel_LIT <= TankEmpty_SP, SET TankFillAuto_Cmd.



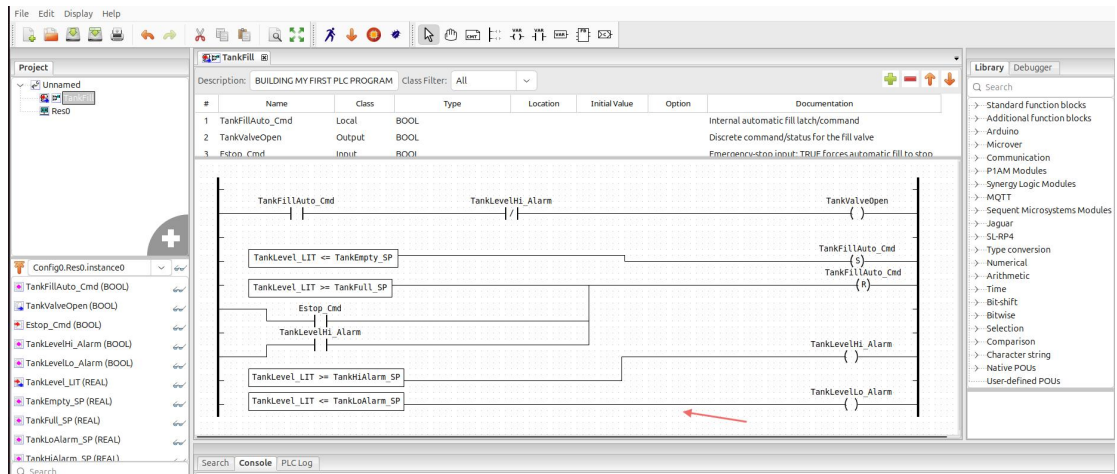
Step 8. Build the automatic-stop/safety rung: when TankLevel_LIT >= TankFull_SP OR Estop_Cmd is TRUE OR TankLevelHi_Alarm is TRUE, RESET TankFillAuto_Cmd.



Step 9. Build the high-level alarm rung: TankLevel_LIT >= TankHiAlarm_SP energizes TankLevelHi_Alarm.



Step 10. Build the low-level alarm rung: TankLevel_LIT <= TankLoAlarm_SP energizes TankLevelLo_Alarm.



Safety explanation

An emergency-stop condition must override normal automatic control because it protects people and equipment regardless of what state the process logic is in. If E-stop only worked when convenient for the sequence, a hazard could occur while the system waited for a normal condition to clear. By wiring Estop_Cmd directly into the RESET branch alongside the high-level and full-tank conditions, the design guarantees that pressing E-stop halts filling immediately and independently of the tank level, satisfying the general safety principle that protective functions take priority over production functions.

Knowledge check answers

1. **Why BOOL and REAL:** BOOL fits values that only have two states, such as commands, alarms, and valve status (on/off). REAL fits continuous physical measurements and setpoints, such as tank level in feet, which vary across a range rather than switching between two states.
2. **Purpose of SET/RESET on TankFillAuto_Cmd:** SET latches the command on when the low setpoint is reached, and RESET clears it when the tank is full, an alarm trips, or E-stop is active. This gives a single command bit that persists

across scans without needing a separate seal-in contact, and makes the two conditions (start filling / stop filling) easy to read as one pair of rungs.

3. **At 9 ft with the valve commanded open:** TankLevel_LIT $\geq$ TankHiAlarm_SP becomes TRUE, so TankLevelHi_Alarm energizes. That same alarm bit is one of the RESET conditions for TankFillAuto_Cmd, so the auto command clears, and TankValveOpen's rung loses its NOT TankLevelHi_Alarm condition, so the valve closes.
4. **Why validate in simulation first:** Simulation lets you find logic errors, wrong setpoints, or missing interlocks without risking a real tank overflowing, equipment damage, or unsafe conditions. Since PLC logic controls physical processes, mistakes caught on a live system can have real consequences, whereas a simulated environment allows safe, repeated testing before connecting to an HMI or real I/O.